

CheatSheet - R basics

Topic	Syntax	Description	Example
R Persistence - Write	<code>saveRDS()</code>	<code>saveRDS</code> function writes a single R object to a file.	<code>saveRDS(movies, "movies.rds")</code>
R Persistence - Read	<code>readRDS()</code>	<code>readRDS</code> function reads the file.	<code>saveRDS(movies, "movies.rds")</code> <code>movies <- readRDS("movies.rds")</code>
Lists	<code>list(num1, num2, num3)</code>	A list is a sequenced collection of different objects of R, like vectors, numbers, characters, other lists as well, and so on. You can consider a list as a container of correlated information, well structured and easy to read.	<code>movie = list("Toy Story", "Akira", "The Breakfast Club")</code>
Access Lists	<code>list_variable_name[]</code>	You can access the list items by referring to its index number, inside brackets. The first item has index 1, the second item has index 2, and so on.	<code>movie<-list("Toy Story", 1995, c("Animation", "Adventure", "Comedy"))</code> <code>movie[2]</code>
Named lists	<code>my_list = list(name1 = your_comp1, name2 = your_comp2)</code>	Named list can also be created using <code>list()</code> function to specify the names of elements after defining the list.	<code>movie = list(name = "Toy Story", year = 1995, genre = c("animation", "adventure", "comedy"))</code>
Adding a new element in list	<code>append(list_name, new_element)</code>	To add an item to the end of the list, use the <code>append()</code> function	<code>append(movie, "horror")</code>
Adding a new element in a named list	<code>list_name[[new_name]]=value</code>	To add an item to the named list, we give the name which will be used as a key or index and the value.	<code>movie[['director']] = "John Lasseter"</code>
Change Item Value in list	<code>my_list[index] = value</code>	To change the value of a specific item, refer to the index number. Index starts from 0.	<code>movie[1] = "Akira"</code>
Remove List Items	<code>my_list = my_list[-index]</code>	You can also remove list items by specifying the index prefixed with minus sign.	<code>movie[-1]</code>
Concatenating lists	<code>mylist_concatenated = c(list1, list2)</code>	Concatenation is the process of putting things together.	<code>movie1<-list("Toy Story", 1995, c("Animation", "Adventure", "Comedy"))</code> <code>movie2<-list("Akira", 1998, c("Animation", "Adventure", "Comedy"))</code> <code>newlist = c(movie1, movie2)</code>
Data Frames	<code>data.frame()</code>	<code>data.frame</code> creates a dataframe with the parameters passed. A dataframe is displayed as a table	<code>movies <- data.frame(name = c("Toy Story", "Akira", "The Breakfast Club", "The Artist", "Modern Times", "Fight Club", "City of God", "The Untouchables"), year = c(1995, 1998, 1985, 2011, 1936, 1999, 2002, 1987), stringsAsFactors=F)</code>
<code>str()</code>	<code>str(object, ...)</code>	<code>str()</code> function is used for compactly displaying the internal structure of a R object as a string.	<code>str(movies)</code>
<code>class()</code>	<code>class(object)</code>	<code>class()</code> function returns the data type of a variable.	<code>class(movies\$year)</code>
<code>head()</code>	<code>head(object)</code>	<code>head()</code> Function returns the first 6 rows of the dataset.	<code>head(movies)</code>
<code>tail()</code>	<code>tail(object)</code>	<code>tail()</code> Function returns the last 6 rows of the dataset.	<code>tail(movies)</code>
<code>rbind()</code>	<code>rbind(object, object, ...)</code>	<code>rbind</code> function combines objects as rows.	<code>rbind(movies, c(name="Dr. Strangelove", year=1964, length=94))</code>

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